Rishabh Baral

3255 South Dorsey Lane #2069, Tempe, AZ 85282 | 214-701-0146 | Contact Me!

U.S. Citizen | Open to AI/ML Research, Data Science Internships, Co-ops, and Full-Time Roles

Education

Arizona State University — Tempe, AZ
Doctor of Philosophy in Computer Science — Expected May 2030

Arizona State University — Tempe, AZ
Master of Science in Computer Science — May 2026

University of Maryland — College Park, MD
Bachelor of Science in Computer Science — May 2024

Research Publications

Moneyball with LLMs: Analyzing Tabular Summarization in Sports Narratives — Jan 2026
R. Upadhyay, R. Baral, N. Ahuja*, V. Gupta | Accepted as Findings Paper to 64th ACL, San Diego, CA; Jul 2-7, 2026*

PHANTOM RECALL: When Familiar Puzzles Fool Smart Models — Jan 2026
S. Mukhopadhyay, R. Baral*, N. Mahajan, et al. | ACL Rolling Review (ARR)*

- Benchmarked LLM reasoning via puzzle variations to expose memorization vs. true reasoning.
- Developed automated error classification achieving 96% human agreement.

"The Moon that Looks Like a Boat": Evaluating Visual Similes in T2I Models — May 2026
Y. Luo, R. Baral*, S. Mukhopadhyay, Y. Yang, C. Baral | ACL Rolling Review (ARR)*

- Evaluated state-of-the-art Text-to-Image (T2I) models on their ability to comprehend and generate visual similes, highlighting current limitations in figurative language alignment.

Technical Skills

Languages: Python, Java, JavaScript, C, C#, Rust, Kotlin, SQL, OCaml, R, MATLAB
ML Libraries: PyTorch, torchvision, Scikit-learn, NumPy, SciPy, Pandas, OpenCV
ML Methods: Computer Vision, Deep Learning, Semi-Supervised Learning, Multi-Task Learning, Transfer Learning
Architectures: ResNet-18, ResNet-50, U-Net, Vision Transformers (ViTs), RAD-DINO
Medical & GIS: DICOM, pydicom, MIMIC-CXR, MedMNIST, TBX11K, QGIS, ArcGIS, GDAL
Tools: CUDA, Git, MongoDB, ImageMagick, VS Code, Android Studio

Experience

Arizona State University — Tempe, AZ
Graduate Project / Teaching Assistant — Aug 2024 – Present

- **Sports Narrative Tabular Summarization:** Contributed to the development and evaluation of LLM-based systems for transforming long-form sports commentary into structured statistical tables. Evaluated GPT-4o and Gemini 2.0 Flash on multi-entity reasoning, numerical consistency, and table generation robustness, leading to a Findings of ACL 2026 publication.
- **Information Assurance:** Performed a comprehensive security analysis for the healthcare industry by applying AI/ML techniques to identify systemic vulnerabilities. Evaluated information assurance protocols and predictive modeling to ensure data integrity and patient privacy within clinical and multimedia database environments.
- **Systems:** Architected and implemented a fully functional Distributed Hash Table (DHT) in Python to manage and retrieve NOAA weather records efficiently. Optimized data distribution, lookup latency, and fault tolerance to handle high-concurrency access to large-scale environmental datasets.
- **Instruction:** Serve as a Computer Science Lab Teaching Assistant, leading instructional sessions on JavaScript for undergraduate students. Responsible for grading technical assignments, troubleshooting full-stack code, and delivering comprehensive review guides to reinforce object-oriented design and programming principles.

Selected Projects

- **Unified 2D/3D Multi-Task Framework for Clinical Radiographs:** Engineered a dimension-agnostic deep learning framework using PyTorch and Axial-Coronal-Sagittal (ACS) convolutions to unify 2D and 3D medical image analysis across 18 MedMNIST v2 datasets. Developed a semi-supervised Student-Teacher pipeline integrating RAD-DINO and Exponential Moving Averages (EMA) to leverage 184,556 unlabeled MIMIC-CXR images, achieving a state-of-the-art Test AUC of 0.878. Implemented an Ark Cyclic Training paradigm within a ResNet-50 backbone to eliminate gradient interference across concurrent classification, localization, and segmentation tasks, while optimizing deployment on Apple M3 silicon to achieve a 2.4× throughput increase. Conducted 20 independent experiments on TBX11K, demonstrating that training from scratch significantly outperformed COCO pre-training in clinical imaging domains.

- **Clinical Data Pipeline & Diagnostic Architectures:** Built a medical imaging pipeline using `pydicom` to parse clinical metadata and normalize high-bit-depth DICOM scans into diagnostic-quality 8-bit representations. Designed a hybrid U-Net segmentation architecture with a ResNet-18 encoder and skip connections to preserve anatomical detail, achieving 98.42% validation pixel accuracy within 10 epochs. Adapted ImageNet-pretrained ResNet-18 models for skeletal age estimation and anatomical bounding box localization through continuous coordinate regression, achieving a localization Mean Absolute Error (MAE) of 0.0909.
- **GIS & Image Processing:** Leveraged QGIS to process and analyze multi-spectral Landsat-8 and Sentinel-2 satellite data, utilizing color composites to identify geographical features. Developed data plots and high-resolution animations of Hurricane Idalia's motion using NOAA GOES-16E imagery and ImageMagick. Conducted advanced spatial analysis through ArcGIS to evaluate the proximity impacts and environmental damage of California wildfires.
- **AI/ML Architecture:** Engineered a custom neural network from scratch, implementing a robust backpropagation algorithm to tune network parameters on synthetic datasets. Developed a comprehensive transfer learning benchmark to evaluate model performance across diverse domains, including computer vision (CV), natural language processing (NLP), and multi-modal integration.
- **MicroCaml:** Architected and engineered "MicroCaml," a functional sublanguage of OCaml. Developed a full pipeline including a custom Lexer for tokenization, a Parser for abstract syntax tree generation, and an Interpreter to execute higher-order functions. Additionally simulated a regular expression engine and a tree-based database system using OCaml.
- **Full-Stack Dev:** Built a responsive, multi-screen weather application in Kotlin featuring real-time API integration, multi-language support, and Google AdMob monetization. Developed a high-concurrency order processing supermarket application using a JavaScript/Node.js backend, a modern HTML/CSS frontend, and a MongoDB database for scalable JSON data storage.